

Supplementary Text

The CSPdb

The CSPdb includes receptors that regulate diverse biological processes, including cell cycle progression (e.g., MAD2: 1KLQ), transcriptional control (e.g., Nrd1: 2MOW, 6GC3), translational regulation, autophagy, apoptosis (e.g., BCL-XL family: 2M04, 2LP8), and cell adhesion (e.g., tensin: 2LOZ). Disease-relevant targets address mechanisms in leukemia and other cancers, Alzheimer’s disease (e.g., amyloid precursor protein-binding proteins: 2ROZ), and infectious diseases from viruses and bacteria, such as anthrax (2RUI), chikungunya (5I22), SARS-CoV-2 (7PKU), HIV (2L6E), and Ebola (2KQ0).

The CSPdb features several recurring protein domains that exhibit notable conformational changes upon ligand binding. For instance, it includes 23 interactions involving calmodulin (CaM) or CaM-like proteins with peptides (e.g., PDB IDs: 1SY9, 2JZI; full list in **SI Table S1**), which often involve substantial rearrangements upon binding described in detail elsewhere¹. Other receptors known for binding-induced conformational shifts or folding include the G protein Cdc42 (1CF4), the metaphase chromosome spindle checkpoint protein MAD2 (1KLQ), the KIX domain (2LXS), and CREB-binding proteins (e.g., 1R8U, 1L8C).

Additional recurring domains include the extra-terminal (ET) domains of bromodomain proteins BRD3 and BRD4 (e.g., 2NCZ, 2ND0; full list in **SI Table S7**, **SI Fig. S6**), which bind intrinsically-disordered regions of viral integrase or chromatin regulator proteins. These interactions trigger subtle changes distal from the binding site, including the formation of a stabilizing secondary structure that reorients two allosteric helices^{2, 3}. Similarly, the PDZ2 domain (e.g., 1D5G, 1VJ6, 2M0V) shows distal effects mediated by the β 2– β 3 loop during peptide recognition⁴.

Nine entries involve the PH domain of the TFIIF transcription initiation factor subunit Tfb1 (e.g., 2LOX, 2M14; full list in **SI Table S8**, **SI Fig. S7**). In these complexes, the ligand is often modeled as a helical polypeptide bound orthogonally to an extended β -sheet, resulting in signal propagation via hydrogen-bond rearrangements. Other targets which involve hydrogen bond rearrangement in β -sheets include the Cdc42 (1CF4), Cyclophilin A (2MS4), and SH3 domain (e.g., 2OJ2, 2ROL; full list in **SI Table S1**). An additional eight entries use ubiquitin or ubiquitin-like (e.g. SUMO and NEDD8) protein receptors. Ubiquitin has been used as a model system for NMR studies⁵ and for studying allostery⁶ (e.g., 2KWV, 2MUR; full list in **SI Table S9**, **SI Fig. S8**).

Estimated Distance Limits for Non-structural NMR Chemical Shift Perturbations

Aromatic ring-current effects decrease as r^{-3} , with established ring-current parameters indicating that a ≥ 0.05 -ppm amide ^1H shift is unlikely beyond approximately 10 Å for Phe/Tyr and 12 Å for Trp, even under idealized geometry^{8, 9}. Similarly, changes in charge or ionization state accompanying ligand binding can produce amide CSPs through changes in the local electric field, although the magnitude of these effects depends strongly on the dielectric environment, screening, and orientation^{10, 11}. Thus, distal CSPs (e.g., > 15 Å from the binding site) generally reflect long-range effects of ligand binding and are consistent with changes in protein conformation or conformational distributions.

NMR ring-current contributions to chemical shift perturbations

Based on the r^{-3} distance dependence of aromatic ring-current effects and established ring-current parameters for aromatic amino acids, the theoretical maximum distance for a ≥ 0.05 -ppm HN ring-current shift is approximately 10 Å for Phe/Tyr and 12 Å for Trp under idealized geometry^{8, 9}.

Estimated maximum distance for a ≥ 0.05 -ppm amide ^1H ring-current shift.

Aromatic ring-current contributions to ^1H chemical shifts can be approximated using the equivalent-dipole

model,

$$\Delta\delta_{\text{RC}} = \mu \frac{(3 \cos^2 \theta - 1)}{r^3},$$

where r is the distance from the proton to the aromatic ring center, θ is the angle between this vector and the normal to the ring plane, and μ is the effective ring-current constant. Abraham et al. determined $\mu = 26.2 \text{ ppm } \text{\AA}^3$ for benzene. The maximum absolute effect occurs at $\theta = 0^\circ$ or 180° corresponding to the proton lying along the normal above or below the aromatic ring plane, where the angular term has a magnitude of 2. Using the 0.05-ppm significance threshold gives

$$r_{\text{max}} = \left(\frac{2\mu}{0.05} \right)^{1/3} = 10.2 \text{ \AA}.$$

Thus, even under idealized geometry, a Phe/Tyr-like aromatic ring would need to be within approximately $10 \text{ \AA}$ of an amide proton to produce a significant ring-current effect. For Trp, the contributions from its two aromatic rings are summed; using the relative ring-current intensities reported by Perkins et al. gives an idealized upper-bound estimate of approximately $12 \text{ \AA}$.

NMR electric-field induced chemical shift perturbations

Based on the r^{-2} distance dependence of electric-field effects and experimentally determined parameters for protein amide chemical shifts, a one-unit charge change is estimated to produce a ≥ 0.05 -ppm HN shift only within approximately $6\text{--}8 \text{ \AA}$ under idealized conditions^{10, 11}.

Estimated maximum distance for a ≥ 0.05 -ppm amide ^1H electric-field induced shift.

Electric-field-induced amide chemical shifts can be approximated using the Buckingham treatment,

$$|\Delta\delta_{\text{HN}}| = |A| |E| |\cos \beta|,$$

where A is the HN shielding polarizability, E is the change in electric field at the amide proton, and β is the angle between these vectors. Using $|A| = 21 \text{ ppm } \text{\AA}^2/e$ for backbone amide protons¹¹ and approximating the electric field from a point-charge change using Coulomb's law gives

$$|\Delta\delta_{\text{HN}}| = \frac{21 |\Delta q|}{\epsilon_{\text{eff}} r^2} |\cos \beta|.$$

For a one-unit charge change ($|\Delta q| = 1e$), optimal orientation ($|\cos \beta| = 1$), and the 0.05-ppm significance threshold,

$$r_{\text{max}} = \left[\frac{21}{0.05 \epsilon_{\text{eff}}} \right]^{1/2}.$$

Experimentally derived effective dielectric constants for proteins are approximately $6\text{--}13$; values of $\sim 6.5\text{--}7$ and 8.6 ± 0.8 have been obtained from protein CSP measurements^{10, 11}. These values correspond to idealized maximum distances of approximately $6\text{--}8 \text{ \AA}$ for a ≥ 0.05 -ppm ^1HN shift from a one-unit charge change.

Unlike aromatic ring-current effects, a universal distance cutoff cannot be assigned to electrostatic CSPs because their magnitude depends on charge, orientation, dielectric environment, and screening. Small electric-field-induced ^1HN CSPs have been experimentally detected $> 12 \text{ \AA}$ from a charge perturbation, although these were below the 0.05-ppm significance threshold used here¹¹. Although our apo and holo datasets were matched for pH, ligand binding itself may alter protein ionization states and thereby generate electric-field-induced CSPs.